

Advanced (Habanero)

- Q1.** If $f(x) = x^2$ and $g(x) = x + 2$, what is $(f + g)(x)$?
(A) $x^2 + x + 2$
(B) $x^2 - x - 2$
(C) $x^2 + 2$
(D) $x + 2$
(E) $x^2 - 2$
- Q2.** What is the domain of $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$ when combined as $(f \cdot g)(x)$?
(A) $(-\infty, 0) \cup (0, 1) \cup (1, \infty)$
(B) $(-\infty, 0) \cup (0, \infty)$
(C) $(-\infty, 1) \cup (1, \infty)$
(D) $(-\infty, 0)$
(E) $(-\infty, 1)$
- Q3.** If $f(x) = x^2 - 2x + 1$ and $g(x) = x - 1$, what is $(f - g)(x)$?
(A) $x^2 - 3x + 2$
(B) $x^2 - x$
(C) $x^2 - 2$
(D) $x - 1$
(E) x^2
- Q4.** If $f(x) = x + 1$ and $g(x) = x - 1$, what is $(f \cdot g)(x)$?
(A) $x^2 - 1$
(B) $x^2 + 1$
(C) $x + 1$
(D) $x - 1$
(E) x
- Q5.** What is the domain of $(f/g)(x)$ when $f(x) = x + 1$ and $g(x) = x - 1$?
(A) $(-\infty, 1) \cup (1, \infty)$
(B) $(-\infty, -1) \cup (1, \infty)$
(C) $(-\infty, 1)$
(D) $(-1, 1)$
(E) $(-\infty, 0)$
- Q6.** If $f(x) = x^2 + 1$ and $g(x) = x^2 - 1$, what is $(f + g)(x)$?
(A) $2x^2$
(B) $2x$
(C) $2x^2 + 2$
(D) x^2
(E) $2x^2 - 2$
- Q7.** If $f(x) = x + 2$ and $g(x) = 2x$, what is $(f \cdot g)(x)$?
(A) $2x^2 + 4x$
(B) $2x + 4$
(C) $2x^2 - 4$
(D) $x + 2$
(E) $2x$
- Q8.** If $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x+1}$, what is $(f - g)(x)$?
(A) $\frac{1}{x(x+1)}$
(B) $\frac{1}{x+1}$
(C) $\frac{x+1}{x}$
(D) $\frac{1}{x}$
(E) $\frac{x}{x+1}$

Open Ended Questions (FRQ)

- Q9.** Find $(f/g)(x)$ when $f(x) = x^2 + 2x - 3$ and $g(x) = x - 1$. Simplify the resulting expression and state the domain.
- Q10.** Given $f(x) = x^2 + 1$ and $g(x) = 2x - 1$, find $(f + g)(x)$ and $(f - g)(x)$. State the domain for each.

Chef's Tips

- Remind students to simplify expressions after performing operations with functions.
- Caution students about the domain when dealing with division of functions.
- Encourage students to use the given functions to derive new functions through operations.